Chemical Engineering Laboratory II

CHE 382

Welcome Back!!

UIC
UNIVERSITY OF ILLINOIS AT CHICAGO

Today’s Schedule

Today we will:

Go over course and lab schedule

Discuss safety in the laboratory

- Read and sign safety document
- Assign Safety reading and quiz
- Assign SACHE safety tutorial

Assign teams and first lab experiments

Chemical Engineering Laboratory II
Room 128 EIB

Section A: Tuesday 9:00AM – 12:30 PM

Section B: Tuesday 1:00PM – 4:30PM

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Teaching Assistants:

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Syllabus

- Full copy available in Blackboard, Course Information folder

CHE382 Experiments

A: Evaporation

B: Liquid-Liquid Extraction

C: UltraFiltration

C: Humidification

E: Gas Absorption

F: Heat Transfer

- Lab manuals and additional handouts are available in Blackboard/Lab Manuals
- **Additional information in BB also available**

Primary Lab Information

- Lab manuals: Most up-to-date versions available through Blackboard

- **Unit Operations in Chemical Engineering textbook**
  - **Great resource for theory and practical information**
  - **USE IT!!**
- Optional: "Writing Style and Standards in Undergraduate Reports", J. Donnell, S. Jeter, C. MacDougall, J. Snedeker, College Publishing, 3rd Edition.
  - See Handout on writing Chemical Engineering reports

Additional Resources

- Additional information is in Blackboard for each experiment
- Perry’s Chemical Engineering Handbook
  - Another great resource for practical info and correlations.
- Online resources/UIC library resources

- NOTE: At least 4 references are required for each final lab report (Not including lab manual)

Lab Schedule Spring 2020 TUESDAY

Week 1:	Jan	14	Introductions, Lab Team assignments	
Week 2:		21	Prelab 1, **Safety Assignment I Due**	**LAB 1**
Week 3:		28	Laboratory 1	
Week 4:	Feb	4	Pre-lab 2	**LAB 2**
Week 5:		11	Laboratory 2	
Week 6:		18	Pre-Lab 3	**LAB 3**
Week 7:		25	Laboratory 3	
Week 8:	Mar	3	Pre-Lab 4	**LAB 4**
Week 9:		10	Laboratory 4	
Week 10:		17	Pre-Lab 5 Safety Quiz II Due 3-20	**LAB 5**
Week 11:		**24**	**SPRING BREAK (No Lab Period)**	
Week 12:		31	Laboratory 5	
Week 13:	Apr	7	Pre-Lab 6	**LAB 6**
Week 14:		14	Laboratory 6	
Week 15:		21	Class presentations (**Final Lab Period)**	
Week 16:		28	**No Lab**	
Week 17:	May	4-8	**FINAL EXAM WEEK (No lab)**	

Lab Schedule Spring 2020 THURSDAY

Week 1:	Jan	16	Introductions, Lab Team assignments	
Week 2:		23	Prelab 1, **Safety Assignment I Due**	**LAB 1**
Week 3:		30	Laboratory 1	
Week 4:	Feb	6	Pre-lab 2	**LAB 2**
Week 5:		13	Laboratory 2	
Week 6:		20	Pre-Lab 3	**LAB 3**
Week 7:		27	Laboratory 3	
Week 8:	Mar	5	Pre-Lab 4	**LAB 4**
Week 9:		12	Laboratory 4	
Week 10:		19	Pre-Lab 5 Safety Quiz II Due 3-20	**LAB 5**
Week 11:		**26**	**SPRING BREAK (No Lab Period)**	
Week 12:	Apr	2	Laboratory 5	
Week 13:		9	Pre-Lab 6	**LAB 6**
Week 14:		16	Laboratory 6	
Week 15:		23	Class presentations (**Final Lab Period)**	
Week 16:		30	**No Lab**	
Week 17:	May	4-8	**FINAL EXAM WEEK (No lab)**	

Course Output

- Pre-Lab Plans
- Final Lab reports
- Oral presentations (every two weeks)
- Safety Assignments:
  - Safety quiz
  - SACHE tutorial
- Final oral presentation (end of semester)

Grade Determination

Pre-Lab Plan	20%
Final Lab Report	40%
Oral Presentations	10%
Safety quiz/ SACHE tutorial	10%
Individual Performance Assessment	20%
	100%

Lab and Assignment Cycle

Two Lab periods for each experiment

- **Period 1 (Lab Prep)**
  - Familiarization with equipment
  - Lab tasks assigned
  - **Perform initial measurements and calibrations**
  - Prepare Pre-Lab Plan with Safety Analysis
- Pre-Lab Plan due at end of lab period

1st Lab Schedule

1st lab will be 3 weeks long; Pre-lab for first lab started today will continue through next week:

Pre-Lab 1 Plan due:

Tuesday January 21st : End of lab section A or B

Thursday January 23rd : End of lab section A or B

Lab and Assignment Cycle

Period 2:

- Devoted to performing the experimental work
- Collecting data, analyzing results, creating graphs, figures and tables, error analysis.
- Write Final Lab Report
- Prepare oral presentation for next lab period

Final Lab Report due:

- Tuesday Labs: Monday at 5:00 PM
- Thursday Labs: Wednesday at 5:00 PM

- **Next Period 1:**
  - Oral presentation to incoming team

Attendance

- Attendance is required for all lab periods.
  - TA will take attendance; please check in before lab starts.
    - On-time: Up to 9:15 AM
    - Late: 9:15 – 9:50 AM
    - Absent: > 9:50 AM
  - If you have to miss a lab you must personally contact instructor or TA by Monday night (email is best). Excused absences may require proof.
  - 3 unexecused absences- Zero on individual assessment grade up to “F” for course grade.

SAFETY

Important Safety Phone Numbers

Campus Emergency

- 5-5555 dialed from local campus phone
- 312-355-5555 from your cell phone

UIC Environmental Health and Safety Office

- **996-SAFE (24/7 emergency number)**

Campus Non-Emergency

- 6-2830 from local campus phone
- 312-996-2830 from your cell phone

Lab Emergency Drill Procedures

- If possible shut down your experiment if lab emergency occurs; otherwise
- Exit building quietly and remain calm
  - Be aware of surroundings and possible hindrances that could occur.
- Wait for further instructions

Specific Safety Considerations

- Corrosive and hazardous chemicals handling
- **<u>Read MSDS sheets</u>** to familiarize yourself with hazards of each chemical and handling guidelines.
- Wear proper safety attire
  - Safety glasses (ALWAYS)
  - Lab coat if needed
  - Safety gloves:
    - Blue nitrile gloves (general labwork)
    - Chemical-resistant gloves (chemical/corrosive hazards)
    - Heat-resistant gloves (hot equipment)
  - Know where the safety shower and eyewash are located

Safety Hazard Examples

- Liquid Extraction Experiment
  - White oil handling and transfer from carboy
  - Glacial acetic acid- use fume hood
  - Standard NaOH solutions (1N, 0.5N, 0.1N, 0.05N 0.01N)
- Ultrafiltration:
  - Phosphoric acid (85%), NaOH pellets, sodium hypochlorite (bleach)
- Gas Absorption:
  - High-pressure CO2 gas cylinder
  - Standard NaOH solutions
- Steam hazard (HOT!): Ultrafiltration, Heat transfer, Evaporation, Humidification

Guidelines on Safety Glasses

- Safety glasses must always be worn on the first floor of the lab.
- Pick up a pair when you enter the laboratory in the morning and put them on.
- Compliance will be monitored by the instructor and TA during the lab period. Non-compliance will be recorded.
  - 3 instances of non-compliance will result in a zero on the Individual Assessment grade.

Safety Assignment #1:
JT Baker Hazardous Chemical Safety

Assignment:

- Review selected chapters from an industrial chemical safety course:
  Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards, Compressed Gases
- Answer the questions in the quiz titled "CHE 382 Safety Quiz". See Blackboard Assignment. Open book/open notes and team discussion are allowed but submit individually to Blackboard.

Grading: Quiz due next week: Tuesday January 21st or Thursday January 23rd by 5pm.

Safety Assignment #2: SACHE-AICHE Safety Tutorial

AICHE (American Institute of Chemical Engineers) has created a website with various safety tutorials covering process safety, risk assessment and chemical reactivity hazards.

ASSIGNMENT: Complete the tutorial titled: **“Dust Explosion Control”** in the website. The procedure on how to access the website will be given by email and also in Blackboard

GRADING: You must submit proof in Blackboard Assignment that you have completed the tutorial (Certificate issued from SACHE) for full credit.

DUE: Friday March 20th 5:00 PM

Lab Experiment Assignment
Read, Sign and Hand In Safety Document

Tuesday AM Project Teams

- 2 teams of 4 members
- 1 team of 5 members

Tuesday PM Project Teams

- 3 teams of 5 members
- 1 team of 4 members

Thursday AM Project Teams

- 2 teams of 5 members

THURSDAY PM Project Teams

- 3 teams of 5 members
- 1 team of 4 members